

TrES-2b

R-1341

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_190607 · created 2026-07-31T21:37:46+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458641.9174 +/- 0.00065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.032
Transit depth (Rp/Rs)^2	2.18 +/- 0.94 [%]
Orbital Inclination (inc)	88.82 +/- 2.21
Airmass coefficient 1 (a1)	1.28 +/- 0.017
Airmass coefficient 2 (a2)	-0.21 +/- 0.19
Scatter in the residuals of the lightcurve fit is	1.34 %
Best Comparison Star	#2 - [257, 336]
Optimal Aperture	3.96
Optimal Annulus	8.24
Transit Duration (day)	0.1074 +/- 0.0076

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.148 $\pm$ 0.032 vs 0.1254 (deviation 0.46 σ)
Duration fit vs published	0.1074 d vs 0.0746 d (deviation 2.79 σ)
Mid-time O-C	-1.5 min
Depth SNR	3.0
Residual scatter	1.34 %

Light curve

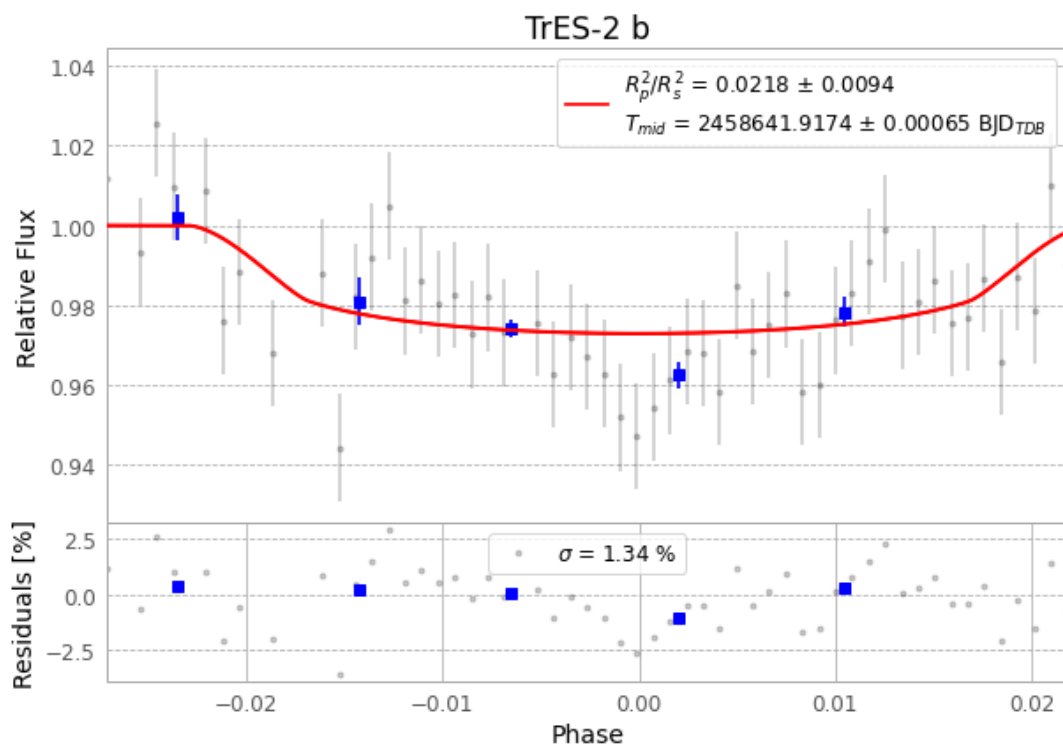


Figure 1 — FinalLightCurve_TrES-2 b_2019-06-07.png (SHA-256 dd33f882261c8908...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [397, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e484eebc3ebcea33...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

59 FITS frames (39 MB), TRES-2190607082112.FITS → TRES-2190607111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-190607.log (84ce1916d41d...), FinalLightCurve_TrES-2 b_2019-06-07.png (dd33f882261c...), FinalLightCurve_TrES-2 b_2019-06-07.csv (8fcc8b33ef69...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 21:37Z.